

The Nexus Knowledge Engine is a RAG system for enterprise knowledge retrieval.

Key Features: - Document processing with PDF text extraction and sliding window chunking - Vector embeddings using sentence-transformers (384 dimensions) - Cosine similarity search via pgvector HNSW index - Redis semantic caching for duplicate queries - Retrieval Gate to block low-confidence responses - MLflow telemetry for query latency and confidence tracking Architecture: FastAPI with asyncpg, PostgreSQL/pgvector, Redis, and MLflow. Documents are uploaded, extracted via PyMuPDF, chunked with